Notes the computation of cross sections related to fragmentation functions in collinear factorisation

1 Efficient computation of the SIDIS cross section

In this section, we will manipulate the formulas for the SIDIS cross section so to make them optimal from the point of view of the implementation. The goal is to make their numerical computation as efficient as possible. The SIDIS differential cross section for the exchange of a virtual photon can be written as:

$$\frac{d^3\sigma}{dxdQdz} = \frac{4\pi\alpha^2}{Q^3}\left[\frac{Y_+}{x}F_2(x,z,Q^2) - \frac{y^2}{x}F_L(x,z,Q^2)\right], \tag{1.1}$$

where we have defined:

$$Y_+ = 1 + (1-y)^2\,. \tag{1.2}$$

The structure functions F_2 and F_L are given at NLO by the following convolution:

$$\begin{aligned} F(x,z,Q) &= x\sum_{q\overline{q}} e_q^2\Big[\left(C_{qq}(x,z|Q)\otimes f_q(x|Q) + C_{qg}(x,z|Q)\otimes f_g(x|Q)\right)\otimes d_q(z|Q) \\ &+ \left(C_{gq}(x,z|Q)\otimes f_q(x|Q)\right)\otimes d_g(z|Q)\Big], \end{aligned} \tag{1.3}$$

where $\{f_q, f_g\}$ are the quark and gluon PDFs and $\{d_q, d_g\}$ are the quark and gluon FFs, e_q is the electric charge of the quark q and $\{C_{qq}^{2,L}, C_{qg}^{2,L}, C_{gq}^{2,L}\}$ are the relevant partonic cross sections. Importantly, the perturbative coefficients of the coefficient functions C factorise as follows:

$$C(x,z) = \sum_t c_t O_t^{(1)}(x) O_t^{(2)}(z)\,, \tag{1.4}$$

where c_t are numerical coefficients, and $O_t^{(1)}$ and $O_t^{(2)}$ are one-dimensional functions of x and z, respectively. Since we are interested in isolating the FFs at some initial scale Q_0, we also write:

$$d_{q(g)}(z|Q) = T_{q(g)i}d_i(z|Q) = T_{q(g)i}\Gamma_{ij}(z|Q,Q_0)\otimes d_j(z|Q_0)\,, \tag{1.5}$$

where Γ_{ij} is the appropriate evolution operator and $T_{q(g)i}$ is the rotation matrix from the evolution to the physical basis. Now, for the generic structure function, let us define:

$$\begin{aligned} \mathbb{C}_j(z|x,Q,Q_0) &= \Bigg[C_{qq}(x,z|Q)\otimes\left(\sum_{q\overline{q}} e_q^2 f_q(x|Q)T_{qi}\right) + \left(\sum_{q\overline{q}} e_q^2 T_{qi}\right) C_{qg}(x,z|Q)\otimes f_g(x|Q) \\ &+ C_{gq}(x,z|Q)\otimes\left(\sum_{q\overline{q}} e_q^2 f_q(x|Q)\right) T_{gi}\Bigg]\otimes\Gamma_{ij}(z|Q,Q_0) \\ &= \mathbb{K}_i^{\rm SIDIS}(z|x,Q)\otimes\Gamma_{ij}(z|Q,Q_0)\,, \end{aligned} \tag{1.6}$$

where we have used the fact that, in the zero-mass scheme, the quark hard cross sections do not depend on the specific quark flavour, such that:

$$F(x,z,Q) = \sum_j \mathbb{C}_j(z|x,Q,Q_0)\otimes d_j(z|Q_0)\,. \tag{1.7}$$

Using the perturbative expansion of the hard cross sections and Eq. (1.4), the factor $\mathbb{K}^{\rm SIDIS}$ in Eq. (1.6) can be written more explicitly as:

$$
\begin{aligned}
\mathbb{K}_i^{\rm SIDIS}(z|x,Q) &= \left(\sum_{q\overline{q}} e_q^2 f_q(x|Q) T_{qi}\right)\delta(1-z)\\
&+ a_s(Q)\sum_t c_{qq,t}\left[O_{qq,t}^{(1)}\otimes\left(\sum_{q\overline{q}} e_q^2 f_q(Q) T_{qi}\right)\right](x) O_{qq,t}^{(2)}(z)\\
&+ a_s(Q)\left(\sum_{q\overline{q}} e_q^2 T_{qi}\right)\sum_t c_{qg,t}\left[O_{qg,t}^{(1)}\otimes f_g(Q)\right](x) O_{qg,t}^{(2)}(z)\\
&+ a_s(Q) T_{gi}\sum_t c_{gq,t}\left[O_{gq,t}^{(1)}\otimes\left(\sum_{q\overline{q}} e_q^2 f_q(Q)\right)\right](x) O_{gq,t}^{(2)}(z)\,,
\end{aligned}
\tag{1.8}
$$

with:

$$
a_s(Q) = \frac{\alpha_s(Q)}{4\pi}\,. \tag{1.9}
$$

Note that, given the evolution and physical basis, one has that $T_{gi}=\delta_{gi}$ and $T_{qg}=0$. Finally, defining:

$$
\mathbb{K}_i^{\rm SIDIS}(z|x,Q) = \frac{4\pi\alpha^2}{Q^3}\left[\frac{Y_+}{x}\mathbb{K}_{2,i}^{\rm SIDIS}(z|x,Q) - \frac{y^2}{x}\mathbb{K}_{L,i}^{\rm SIDIS}(z|x,Q)\right]\,, \tag{1.10}
$$

one has:

$$
\frac{d^3\sigma}{dxdQdz} = \sum_j \mathbb{K}_i^{\rm SIDIS}(z|x,Q)\otimes\Gamma_{ij}(z|Q,Q_0)\otimes d_j(z|Q_0)\,. \tag{1.11}
$$

1.1 NNLO expressions

Extending the calculation of the SIDIS cross section to NNLO accuracy implies, on top of the mere inclusion of the $\mathcal{O}(\alpha_s^2)$ contribution to the hard cross sections, a number of complications. First, the flavour structure in Eq. (1.3) becomes significantly more involved. Second, the factorisation of the hard cross sections in terms of bi-linear operators in Eq. (1.4) no longer holds. In this section, we will show how to overcome these complications.

1.1.1 Electromagnetic structure functions

We use the notation of Refs. [1, 2] focusing on the flavour structure. Therefore, we neglect all dependences and Mellin convolutions. Moreover, here we concentrate on electromagnetic structure functions, *i.e.* those structure function characterised by the exchange of virtual photon. This has the consequence that only electric charges will appear. The following discussion is applicable to both unpolarised and polarised structure functions, which, under the assumption of photon-exchange only, are F_2 and F_L in the unpolarised case, and g_1 in the polarised case. With these premises, the structure of a SIDIS structure functions to NNLO accuracy reads:

$$
\begin{aligned}
F &= C_{qq}^{\rm NS}\left(\sum_\alpha e_\alpha^2 f_\alpha d_\alpha\right) + C_{qg} f_g\left(\sum_\alpha e_\alpha^2 d_\alpha\right) + C_{gq} d_g\left(\sum_\beta e_\beta^2 f_\beta\right)\\
&+ C_{gg}\left(\sum_\gamma e_\gamma^2\right) f_g d_g + C_{qq}^{\rm PS}\left(\sum_\gamma e_\gamma^2\right)\left(\sum_\alpha f_\alpha d_\alpha\right) + C_{\overline{q}q}\left(\sum_\alpha e_\alpha^2 f_{-\alpha} d_\alpha\right)\\
&+ C_{q'q}^{(1)}\left(\sum_{\substack{\alpha,\beta\\ \alpha\neq\beta,-\beta}} e_\beta^2 f_\beta d_\alpha\right) + C_{q'q}^{(2)}\left(\sum_{\substack{\alpha,\beta\\ \alpha\neq\beta,-\beta}} e_\alpha^2 f_\beta d_\alpha\right) + C_{q'q}^{(3)}\left(\sum_{\substack{\alpha,\beta\\ \alpha\neq\beta,-\beta}} e_\alpha e_\beta f_\beta d_\alpha\right)\,,
\end{aligned}
\tag{1.12}
$$

where the second and third lines only appear at NNLO. Moreover, the coefficient functions C can no longer be written as in Eq. (1.4). The indices α, β, and γ run over all active quark flavours *and* anti-flavours, *i.e.* they are indices in the physical basis, and in principle also over the gluon. We now use Eq. (1.5) to rotate FFs from the physical to the evolution basis and to introduce their evolution. We rewrite Eq. (1.5) as follows

$$d_\alpha = T_{\alpha i} d_i = T_{\alpha i}\Gamma_{ij} d_j^{(0)}\,, \tag{1.13}$$

where we removed all dependences and convolution symbols and, with abuse of notation, we indicated with d_i, or with any latin index, the i-th FF in the evolution basis, while $d_j^{(0)}$ is the j-th FF in the evolution basis at the initial scale Q_0. Also note that $T_{gi} = \delta_{gi}$ and $T_{g\alpha} = \delta_{g\alpha}$. We now use this relation to replace all FFs in the physical basis with their evolution basis counterparts. We then have:

$$F = \mathbb{K}_i \Gamma_{ij} d_j^{(0)}\,, \tag{1.14}$$

where now:

$$\begin{aligned}
\mathbb{K}_i \;=\;& C_{qq}^{\rm NS}\left(\sum_\alpha e_\alpha^2 f_\alpha T_{\alpha i}\right) + C_{qg} f_g \left(\sum_\alpha e_\alpha^2 T_{\alpha i}\right) + C_{gq} T_{gi}\left(\sum_\beta e_\beta^2 f_\beta\right)\\
+\;& C_{gg}\left(\sum_\gamma e_\gamma^2\right) f_g T_{gi} + C_{qq}^{\rm PS}\left(\sum_\gamma e_\gamma^2\right)\left(\sum_\alpha f_\alpha T_{\alpha i}\right) + C_{\overline{q}q}\left(\sum_\alpha e_\alpha^2 f_{-\alpha} T_{\alpha i}\right)\\
+\;& C_{q'q}^{(1)}\left(\sum_{\substack{\alpha,\beta\\ \alpha\neq\beta,-\beta}} e_\beta^2 f_\beta T_{\alpha i}\right) + C_{q'q}^{(2)}\left(\sum_{\substack{\alpha,\beta\\ \alpha\neq\beta,-\beta}} e_\alpha^2 f_\beta T_{\alpha i}\right) + C_{q'q}^{(3)}\left(\sum_{\substack{\alpha,\beta\\ \alpha\neq\beta,-\beta}} e_\alpha e_\beta f_\beta T_{\alpha i}\right)\,,
\end{aligned} \tag{1.15}$$

where, importantly, greek indices (α, β, and γ) only run over active quarks and antiquarks and not over the gluon. Given the structure of $T_{\alpha i}$, this has the consequence that if i corresponds to the gluon only the terms proportional to C_{gq} and C_{gg} survive. Conversely, if i corresponds to any of the quark quark combinations, the terms proportional to C_{gq} and C_{gg} drop while all others remain. Therefore, we could write:

$$\mathbb{K}_g = C_{gq}\left(\sum_\beta e_\beta^2 f_\beta\right) + C_{gg}\left(\sum_\gamma e_\gamma^2\right) f_g\,, \tag{1.16}$$

while:

$$\begin{aligned}
\mathbb{K}_{i\neq g} \;=\;& C_{qq}^{\rm NS}\left(\sum_\alpha e_\alpha^2 f_\alpha T_{\alpha i}\right) + C_{qg} f_g \left(\sum_\alpha e_\alpha^2 T_{\alpha i}\right)\\
+\;& C_{qq}^{\rm PS}\left(\sum_\gamma e_\gamma^2\right)\left(\sum_\alpha f_\alpha T_{\alpha i}\right) + C_{\overline{q}q}\left(\sum_\alpha e_\alpha^2 f_{-\alpha} T_{\alpha i}\right)\\
+\;& C_{q'q}^{(1)}\left(\sum_{\substack{\alpha,\beta\\ \alpha\neq\beta,-\beta}} e_\beta^2 f_\beta T_{\alpha i}\right) + C_{q'q}^{(2)}\left(\sum_{\substack{\alpha,\beta\\ \alpha\neq\beta,-\beta}} e_\alpha^2 f_\beta T_{\alpha i}\right) + C_{q'q}^{(3)}\left(\sum_{\substack{\alpha,\beta\\ \alpha\neq\beta,-\beta}} e_\alpha e_\beta f_\beta T_{\alpha i}\right)\,.
\end{aligned} \tag{1.17}$$

Since this time it is convenient to parameterise the cross section in terms of the structure functions F_T and F_L, with $F_L = F_2 - F_T$, we have that:

$$\mathbb{K}_i^{\rm SIDIS} = \frac{8\pi\alpha^2}{Q^3}\left[\frac{Y_+}{2x}\mathbb{K}_{T,i} + \frac{1-y}{x}\mathbb{K}_{L,i}\right]\,. \tag{1.18}$$

1.1.2 Electroweak structure functions

In this section, we generalise the discussion of the previous section by considering, on top of photon exchange, also Z and W exchange. This leads to several complications:

- The need to define neutral-current (NC) and charged-current (CC) structure functions characterised by the exchange of neutral γ/Z boson and a W, respectively.
- The introduction of Q-dependent vector and axial electroweak couplings,
- The need to introduce, in the unpolarised case, the parity-violating structure function F_3.

In what follows, we will adhere to the notation of Ref. [3]. We aim at computing the structure function $\mathcal{F}_i^{jk}$, with $i = T, L, 3$ and $jk = \gamma\gamma, \gamma Z, ZZ, W^+, W^-$, such that we can define NC and CC structure functions respectively as the following combinations:

$$
\begin{aligned}
\mathcal{F}_i^{\rm NC} &= \mathcal{F}_i^{\gamma\gamma} - g_V^\ell P_Z \mathcal{F}_i^{\gamma Z} + (g_V^{\ell 2} + g_A^{\ell 2}) P_Z^2 \mathcal{F}_i^{\gamma Z}\,, \quad i = T, L \\
\mathcal{F}_3^{\rm NC} &= -g_A^\ell P_Z \mathcal{F}_3^{\gamma Z} + 2 g_V^\ell g_A^\ell P_Z^2 \mathcal{F}_3^{\gamma Z}\,, \\
\mathcal{F}_i^{\rm CC} &= \mathcal{F}_i^{W^\pm}\,,
\end{aligned}
\tag{1.19}
$$

where we assumed an unpolarised lepton beam, and used the definitions:

$$
g_V^\ell = -\frac{1}{2} + 2\sin^2\theta_{\rm W}\,, \quad g_A^\ell = -\frac{1}{2}\,, \quad P_Z = \frac{G_F M_Z^2}{2\sqrt{2}\alpha}\frac{Q^2}{Q^2 + M_Z^2} = \frac{1}{4\sin^2\theta_{\rm W}(1-\sin^2\theta_{\rm W})}\frac{Q^2}{Q^2+M_Z^2}\,. \tag{1.20}
$$

The general structure of the structure functions reads as follows:

$$
\begin{aligned}
\mathcal{F}_i^{jk} &= \sum_\alpha \mathcal{C}_{qq}^{i,jk}(\alpha) d_\alpha f_\alpha + \sum_\beta \mathcal{C}_{gq}^{i,jk}(\beta) d_g f_\beta + \sum_\alpha \mathcal{C}_{qg}^{i,jk}(\alpha) d_\alpha f_g + \mathcal{C}_{gg}^{i,jk} d_g f_g + \sum_\alpha \mathcal{C}_{\overline{q}q}^{i,jk}(\alpha) d_{\overline{\alpha}} f_\alpha \\
&+ \sum_{\substack{\alpha,\beta,\alpha\beta>0\\ \alpha\neq\beta}} \mathcal{C}_{q'q}^{i,jk}(\alpha,\beta) d_\alpha f_\beta + \sum_{\substack{\alpha,\beta,\alpha\beta<0\\ \alpha\neq-\beta}} \mathcal{C}_{\overline{q}'q}^{i,jk}(\alpha,\beta) d_\alpha f_\beta \\
&= \sum_\alpha \left[C_{qq}^{i,jk}(\alpha) + C_{qq,\rm Fcon}^{i,jk}(\alpha) + C_{qq,\rm NoW}^{i,jk}(\alpha) + C_{q'q}^{i,jk}(\alpha,\alpha) + C_{q'q,\rm NoW}^{i,jk}(\alpha,\alpha)\right] d_\alpha f_\alpha \\
&+ \sum_\beta \left[C_{gq}^{i,jk}(\beta) + C_{gq,\rm NoW}^{i,jk}(\beta)\right] d_g f_\beta + \sum_\alpha \left[C_{qg}^{i,jk}(\alpha) + C_{qg,\rm NoW}^{i,jk}(\alpha)\right] d_\alpha f_g \\
&+ \left[C_{gg}^{i,jk} + C_{gg,\rm NoW}^{i,jk}\right] d_g f_g + \sum_\alpha \left[C_{\overline{q}q}^{i,jk}(\alpha) + C_{\overline{q}q,\rm Fcon}^{i,jk}(\alpha) + C_{\overline{q}'q}^{i,jk}(\overline{\alpha},\alpha) + C_{\overline{q}'q,\rm NoW}^{i,jk}(\overline{\alpha},\alpha)\right] d_{\overline{\alpha}} f_\alpha \\
&+ \sum_{\substack{\alpha,\beta,\alpha\beta>0\\ \alpha\neq\beta}} \left[C_{q'q}^{i,jk}(\alpha,\beta) + C_{q'q,\rm NoW}^{i,jk}(\alpha,\beta)\right] d_\alpha f_\beta + \sum_{\substack{\alpha,\beta,\alpha\beta<0\\ \alpha\neq-\beta}} \left[C_{\overline{q}'q}^{i,jk}(\overline{\alpha},\beta) + C_{\overline{q}'q,\rm NoW}^{i,jk}(\overline{\alpha},\beta)\right] d_\alpha f_\beta\,,
\end{aligned}
\tag{1.21}
$$

where $jk = \gamma\gamma, \gamma Z, ZZ$ and the indices α and β run over quark *and* antiquark. Moreover, we used the notation $\overline{\alpha} = -\alpha$ and $\overline{\beta} = -\beta$. The dependence on the flavour indices of the coefficient functions C for each component jk factors out in terms of appropriate charges. Sketchily, $C_{\dots}^{i,jk}(\alpha) = A_\alpha^{i,jk} C_{\dots,\dots}^{i}$ and $C_{\dots}^{i,jk}(\alpha,\beta) = A_{\alpha\beta}^{i,jk} C_{\dots,\dots}^{i}$.

Let us first consider the neutral-current case for $i = T, L$ which is given by:

$$
\mathcal{F}_i^{\rm NC} = \sum_{jk=\gamma\gamma,\gamma Z,ZZ} \eta_{jk} \mathcal{F}_i^{jk}\,, \tag{1.22}
$$

where we have:

$$
\eta_{\gamma\gamma} = 1\,, \quad \eta_{\gamma Z} = -g_V^\ell P_Z\,, \quad \eta_{ZZ} = (g_V^{\ell 2} + g_A^{\ell 2}) P_Z^2\,. \tag{1.23}
$$

Given the up- and down-type quark (but not antiquark) couplings:

$$
g_V^{d,s,b} = -\frac{1}{2} + \frac{2}{3}\sin^2\theta_{\rm W}\,, \quad g_V^{u,c,t} = \frac{1}{2} - \frac{4}{3}\sin^2\theta_{\rm W}\,, \quad g_A^{d,s,b} = -\frac{1}{2}\,, \quad g_A^{u,c,t} = \frac{1}{2}\,, \tag{1.24}
$$

which for antiquarks are $g_V^{-\alpha} = -g_V^\alpha$ (like the electric change $e_{-\alpha} = -e_\alpha$) and $g_A^{-\alpha} = g_A^\alpha$, the purpose is to compute:

$$
\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \eta_{jk} C_{\dots}^{i,jk}(\alpha,\dots)\,. \tag{1.25}
$$

To this end, we introduce the charges:

$$
\begin{aligned}
A_{\alpha\beta} &= (g_V^{\ell 2} + g_A^{\ell 2}) g_A^\alpha g_A^\beta P_Z^2 \,, \\
B_{\alpha\beta} &= e_\alpha e_\beta - g_V^\ell (e_\alpha g_V^\beta + e_\beta g_V^\alpha) P_Z + (g_V^{\ell 2} + g_A^{\ell 2}) g_V^\alpha g_V^\beta P_Z^2 + A_{\alpha\beta} \,,
\end{aligned}
\tag{1.26}
$$

which produce:

$$
\begin{aligned}
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \eta_{jk} C_{qq}^{i,jk}(\alpha) = B_{\alpha\alpha} C_{qq}^i \,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \eta_{jk} C_{qq,\mathrm{Fcon}}^{i,jk}(\alpha) = B_{\alpha\alpha} C_{qq,\mathrm{Fcon},1}^i + \left(\sum_\gamma B_{\gamma\gamma}\right) C_{qq,\mathrm{Fcon},2}^i \,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \eta_{jk} C_{qq,\mathrm{NoW}}^{i,jk}(\alpha) = \left(\sum_\gamma A_{\alpha\gamma}\right) C_{qq,\mathrm{A},\mathrm{NoW}}^i \,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \eta_{jk} C_{\overline{q}q}^{i,jk}(\alpha) = B_{\alpha\alpha} C_{\overline{q}q}^i \,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \eta_{jk} C_{\overline{q}q,\mathrm{Fcon}}^{i,jk}(\alpha) = B_{\alpha\alpha} C_{\overline{q}q,\mathrm{Fcon}}^i \,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \eta_{jk} C_{q'q}^{i,jk}(\alpha,\beta) = B_{\beta\beta} C_{q'q,1}^i + B_{\alpha\alpha} C_{q'q,2}^i \,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \eta_{jk} C_{q'q,\mathrm{NoW}}^{i,jk}(\alpha,\beta) = (B_{\alpha\beta} - A_{\alpha\beta}) C_{q'q,\mathrm{NoW},3}^i + A_{\alpha\beta} C_{q'q,\mathrm{NoW},4}^i \,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \eta_{jk} C_{\overline{q}'q}^{i,jk}(\alpha,\beta) = B_{\beta\beta} C_{q'q,1}^i + B_{\alpha\alpha} C_{q'q,2}^i \,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \eta_{jk} C_{\overline{q}'q,\mathrm{NoW}}^{i,jk}(\alpha,\beta) = (-B_{\alpha\beta} + A_{\alpha\beta}) C_{q'q,\mathrm{NoW},3}^i + A_{\alpha\beta} C_{q'q,\mathrm{NoW},4}^i \,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \eta_{jk} C_{gq}^{i,jk}(\alpha) = B_{\alpha\alpha} C_{gq}^i \,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \eta_{jk} C_{gq,\mathrm{NoW}}^{i,jk}(\alpha) = \left(\sum_\gamma A_{\alpha\gamma}\right) C_{gq,\mathrm{A},\mathrm{NoW}}^i \,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \eta_{jk} C_{qg}^{i,jk}(\alpha) = B_{\alpha\alpha} C_{qg}^i \,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \eta_{jk} C_{qg,\mathrm{NoW}}^{i,jk}(\alpha) = \left(\sum_\gamma A_{\alpha\gamma}\right) C_{qg,\mathrm{A},\mathrm{NoW}}^i \,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \eta_{jk} C_{gg}^{i,jk} = \sum_\gamma B_{\gamma\gamma} C_{gg}^i \,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \eta_{jk} C_{gg,\mathrm{NoW}}^{i,jk} = \left(\sum_{\gamma,\delta} A_{\gamma\delta}\right) C_{gg,\mathrm{AA},\mathrm{NoW}}^i \,.
\end{aligned}
\tag{1.27}
$$

Now, we can plug these expressions into Eq. (1.22), which finally gives:

$$
\begin{aligned}
\mathcal{F}_i^{\mathrm{NC}} &= C_{qq}^{\mathrm{NS},i}\sum_\alpha B_{\alpha\alpha}d_\alpha f_\alpha + C_{gq}^i\sum_\beta B_{\alpha\alpha}d_g f_\beta + C_{qg}^i\sum_\alpha B_{\alpha\alpha}d_\alpha f_g \\
&+ C_{gg}^i\left(\sum_\gamma B_{\gamma\gamma}\right)d_g f_g + C_{qq}^{\mathrm{PS},i}\left(\sum_\gamma B_{\gamma\gamma}\right)\sum_\alpha d_\alpha f_\alpha + C_{\overline{q}q}^i\sum_\alpha B_{\alpha\alpha}d_{\overline{\alpha}}f_\alpha \\
&+ C_{q'q}^{(1),i}\sum_{\substack{\alpha,\beta\\ \alpha\neq\beta,-\beta}} B_{\beta\beta}d_\alpha f_\beta + C_{q'q}^{(2),i}\sum_{\substack{\alpha,\beta\\ \alpha\neq\beta,-\beta}} B_{\alpha\alpha}d_\alpha f_\beta + C_{q'q}^{(3),i}\sum_{\substack{\alpha,\beta\\ \alpha\neq\beta,-\beta}} B_{\alpha\beta}d_\alpha f_\beta \\
&+ \overline{C}_{qq}^i\sum_\alpha\left(\sum_\gamma A_{\alpha\gamma}\right)d_\alpha f_\alpha + \overline{C}_{gq}^i\sum_\beta\left(\sum_\gamma A_{\beta\gamma}\right)d_g f_\beta + \overline{C}_{qg}^i\sum_\alpha\left(\sum_\gamma A_{\alpha\gamma}\right)d_\alpha f_g \\
&+ \overline{C}_{gg}^i\left(\sum_{\gamma,\delta} A_{\gamma\delta}\right)d_g f_g - C_{q'q}^{(3),i}\sum_{\substack{\alpha,\beta\\ \alpha\beta>0}} A_{\alpha\beta}d_\alpha(f_\beta - f_{\overline{\beta}}) + C_{q'q}^{(4),i}\sum_{\substack{\alpha,\beta\\ \alpha\beta>0}} A_{\alpha\beta}d_\alpha(f_\beta + f_{\overline{\beta}})\,,
\end{aligned}
\tag{1.28}
$$

where, to align the notation with the electromagnetic case discussed in the previous section, we defined:

$$
\begin{aligned}
&C_{qq}^{\mathrm{NS},i} = C_{qq}^i + C_{qq,\mathrm{Fcon},1}^i + C_{q'q,1}^i + C_{q'q,2}^i + C_{q'q,\mathrm{NoW},3}^i\,,\\
&C_{qq}^{\mathrm{PS},i} = C_{qq,\mathrm{Fcon},2}^i\,,\\
&C_{\overline{q}q}^i = C_{\overline{q}q}^i + C_{\overline{q}q,\mathrm{Fcon}}^i + C_{q'q,1}^i + C_{q'q,2}^i + C_{q'q,\mathrm{NoW},3}^i\,,\\
&C_{q'q}^{(1),i} = C_{q'q,1}^i\,,\\
&C_{q'q}^{(2),i} = C_{q'q,2}^i\,,\\
&C_{q'q}^{(3),i} = C_{q'q,\mathrm{NoW},3}^i\,,\\
&C_{q'q}^{(4),i} = C_{q'q,\mathrm{NoW},4}^i\,,\\
&\overline{C}_{qq}^i = C_{qq,\mathrm{A,NoW}}^i\,,\\
&\overline{C}_{gq}^i = C_{gq,\mathrm{A,NoW}}^i\,,\\
&\overline{C}_{qg}^i = C_{qg,\mathrm{A,NoW}}^i\,,\\
&\overline{C}_{gg}^i = C_{gg,\mathrm{AA,NoW}}^i\,.
\end{aligned}
\tag{1.29}
$$

We can now consider the parity violating structure function $\mathcal{F}_3^{\mathrm{NC}}$, which is given by:

$$
\mathcal{F}_3^{\mathrm{NC}} = \sum_{jk=\gamma\gamma,\gamma Z,ZZ}\xi_{jk}\mathcal{F}_3^{jk}\,. \tag{1.30}
$$

with:

$$
\xi_{\gamma\gamma} = 0\,,\quad \xi_{\gamma Z} = -g_A^\ell P_Z\,,\quad \xi_{ZZ} = 2g_V^\ell g_A^\ell P_Z^2\,, \tag{1.31}
$$

We then introduce the charges:

$$
\begin{aligned}
&D_{\alpha\beta} = -2g_A^\ell P_Z e_\alpha g_A^\beta + 2g_V^\ell g_A^\ell P_Z^2(g_A^\alpha g_V^\beta + g_V^\alpha g_A^\beta)\,,\\
&E_{\alpha\beta} = -2g_A^\ell P_Z g_A^\alpha e_\beta + 4g_V^\ell g_A^\ell P_Z^2 g_A^\alpha g_V^\beta\,,\\
&E_\beta = \sum_\alpha E_{\alpha\beta}\,.
\end{aligned}
\tag{1.32}
$$

With these definitions at hand, we have:

$$
\begin{aligned}
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \xi_{jk} C_{qq}^{3,jk}(\alpha) = D_{\alpha\alpha} C_{qq}^{3}\,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \xi_{jk} C_{qq,\mathrm{Fcon}}^{3,jk}(\alpha) = D_{\alpha\alpha} C_{qq,\mathrm{Fcon},1}^{3}\,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \xi_{jk} C_{qq,\mathrm{NoW}}^{3,jk}(\alpha) = E_{\alpha} C_{qq,\mathrm{A},\mathrm{NoW}}^{3}\,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \xi_{jk} C_{\overline{q}q}^{3,jk}(\alpha) = D_{\alpha\alpha} C_{\overline{q}q}^{3}\,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \xi_{jk} C_{\overline{q}q,\mathrm{Fcon}}^{3,jk}(\alpha) = D_{\alpha\alpha} C_{\overline{q}q,\mathrm{Fcon}}^{3}\,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \xi_{jk} C_{q'q}^{3,jk}(\alpha,\beta) = D_{\beta\beta} C_{q'q,1}^{3} + D_{\alpha\alpha} C_{q'q,2}^{3}\,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \xi_{jk} C_{q'q,\mathrm{NoW}}^{3,jk}(\alpha,\beta) = D_{\beta\alpha} C_{q'q,\mathrm{NoW},3}^{3} + D_{\alpha\beta} C_{q'q,\mathrm{NoW},4}^{3}\,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \xi_{jk} C_{\overline{q}'q}^{3,jk}(\alpha,\beta) = D_{\beta\beta} C_{q'q,1}^{3} - D_{\alpha\alpha} C_{q'q,2}^{3}\,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \xi_{jk} C_{\overline{q}'q,\mathrm{NoW}}^{3,jk}(\alpha,\beta) = D_{\beta\alpha} C_{q'q,\mathrm{NoW},3}^{3} - D_{\alpha\beta} C_{q'q,\mathrm{NoW},4}^{3}\,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \xi_{jk} C_{gq}^{3,jk}(\alpha) = D_{\alpha\alpha} C_{gq}^{3}\,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \xi_{jk} C_{gq,\mathrm{NoW}}^{3,jk}(\alpha) = E_{\alpha} C_{gq,\mathrm{A},\mathrm{NoW}}^{3}\,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \xi_{jk} C_{qg}^{3,jk}(\alpha) = D_{\alpha\alpha} C_{qg}^{3}\,, \\
&\sum_{jk=\gamma\gamma,\gamma Z,ZZ} \xi_{jk} C_{qg,\mathrm{NoW}}^{3,jk}(\alpha) = E_{\alpha} C_{qg,\mathrm{A},\mathrm{NoW}}^{3}\,,
\end{aligned}
\tag{1.33}
$$

which produces:

$$
\begin{aligned}
\mathcal{F}_3^{\mathrm{NC}} &= C_{qq}^{\mathrm{NS},3} \sum_{\alpha} D_{\alpha\alpha} d_\alpha f_\alpha + C_{gq}^{3} \sum_{\beta} D_{\beta\beta} d_g f_\beta + C_{qg}^{3} \sum_{\alpha} D_{\alpha\alpha} d_\alpha f_g \\
&+ C_{\overline{q}q}^{3} \sum_{\alpha} D_{\alpha\alpha} d_{\overline{\alpha}} f_\alpha + C_{q'q}^{(1),3} \sum_{\substack{\alpha,\beta \\ \alpha\neq\beta,-\beta}} D_{\beta\beta} d_\alpha f_\beta + C_{q'q}^{(2),3} \sum_{\substack{\alpha,\beta \\ \alpha\neq\beta,-\beta}} D_{\alpha\alpha} d_\alpha f_\beta \\
&+ C_{q'q}^{(3),3} \left[\sum_{\alpha,\beta,\alpha\beta>0} (D_{\beta\alpha} d_\alpha + D_{\beta\alpha} d_{\overline{\alpha}}) f_\beta \right] + C_{q'q}^{(4),3} \left[\sum_{\alpha,\beta,\alpha\beta>0} (D_{\alpha\beta} d_\alpha - D_{\alpha\beta} d_{\overline{\alpha}}) f_\beta \right] \\
&+ \overline{C}_{qq}^{3} \sum_{\alpha} E_\alpha d_\alpha f_\alpha + \overline{C}_{gq}^{3} \sum_{\beta} E_\beta d_g f_\beta + \overline{C}_{qg}^{3} \sum_{\alpha} E_\alpha d_\alpha f_g\,,
\end{aligned}
\tag{1.34}
$$

where we have defined:

$$
\begin{aligned}
C^{\mathrm{NS},3}_{qq} &= C^3_{qq} + C^3_{qq,\mathrm{Fcon},1} + C^3_{q'q,1} + C^3_{q'q,2}\,,\\
C^3_{\overline{q}q} &= C^3_{\overline{q}q} + C^3_{\overline{q}q,\mathrm{Fcon}} + C^3_{q'q,1} + C^3_{q'q,2}\,,\\
C^{(1),3}_{q'q} &= C^3_{q'q,1}\,,\\
C^{(2),3}_{q'q} &= C^3_{q'q,2}\,,\\
C^{(3),3}_{q'q} &= C^3_{q'q,\mathrm{NoW},3}\,,\\
C^{(4),3}_{q'q} &= C^3_{q'q,\mathrm{NoW},4}\,,\\
\overline{C}^3_{qq} &= C^3_{qq,\mathrm{A},\mathrm{NoW}}\,,\\
\overline{C}^3_{gq} &= C^3_{gq,\mathrm{A},\mathrm{NoW}}\,,\\
\overline{C}^3_{qg} &= C^3_{qg,\mathrm{A},\mathrm{NoW}}\,.
\end{aligned}
\tag{1.35}
$$

We can now move to the CC structure functions characterised by the exchange of a virtual W boson, *i.e* $jk = W^+W^-$. As compared to the NC case, the "NoW" terms appearing in Eq. (1.21) can be dropped straight away, which significantly simplifies the general structure. However, contrary to the NC case, CC interactions are not flavour diagonal, but they are instead flavour changing. More specifically, for any quark up(down)-type quark interacting with the W boson, any down(up)-type quark can be produced. The strength of the coupling is determined by the elements of the 3×3 CKM matrix $V_{\alpha\beta}$, with $\alpha \in U = \{u,c,t\}$ and $\beta \in D = \{d,s,b\}$, with $Q = D\cup U$. Considering the case $i = T, L$ for the exchange of W^-, one finds:

$$
\begin{aligned}
\frac{1}{2}\mathcal{F}_i^{W^-} = \;& C^i_{qq}\sum_{\substack{\alpha\in U\\ \beta\in D}}|V_{\alpha\beta}|^2(d_\beta f_\alpha + d_{\overline{\alpha}} f_{\overline{\beta}}) + C^i_{gq}\sum_{\substack{\alpha\in U\\ \beta\in D}}|V_{\alpha\beta}|^2 d_g(f_\alpha + f_{\overline{\beta}}) + C^i_{qg}\sum_{\substack{\alpha\in U\\ \beta\in D}}|V_{\alpha\beta}|^2(d_\beta + d_{\overline{\alpha}})f_g\\
+\;& C^i_{gg}\sum_{\substack{\alpha\in U\\ \beta\in D}}|V_{\alpha\beta}|^2 d_g f_g + C^i_{qq,\mathrm{Fcon},2}\sum_{\substack{\alpha\in U\\ \beta\in D}}|V_{\alpha\beta}|^2\sum_{f\in Q}(d_f f_f + d_{\overline{f}} f_{\overline{f}}) + C^i_{\overline{q}q}\sum_{\substack{\alpha\in U\\ \beta\in D}}|V_{\alpha\beta}|^2(d_{\overline{\beta}} f_\alpha + d_\alpha f_{\overline{\beta}})\\
+\;& C^i_{qq,\mathrm{Fcon},1}\sum_{\substack{\alpha\in U\\ \beta\in D}}|V_{\alpha\beta}|^2(d_\alpha f_\alpha + d_{\overline{\beta}} f_{\overline{\beta}}) + C^i_{\overline{q}q,\mathrm{Fcon}}\sum_{\substack{\alpha\in U\\ \beta\in D}}|V_{\alpha\beta}|^2(d_{\overline{\alpha}} f_\alpha + d_\beta f_{\overline{\beta}})\\
+\;& C^i_{q'q,1}\sum_{f\in Q}\sum_{\substack{\alpha\in U\\ \beta\in D}}|V_{\alpha\beta}|^2(d_f + d_{\overline{f}})(f_\alpha + f_{\overline{\beta}}) + C^i_{q'q,2}\sum_{f\in Q}\sum_{\substack{\alpha\in U\\ \beta\in D}}|V_{\alpha\beta}|^2(d_\beta + d_{\overline{\alpha}})(f_f + f_{\overline{f}})\,.
\end{aligned}
\tag{1.36}
$$

The exchange of a W^+ is easily obtained from the expression above by exchanging $U \leftrightarrow D$, that is:

$$
\begin{aligned}
\frac{1}{2}\mathcal{F}_i^{W^+} = \;& C^i_{qq}\sum_{\substack{\alpha\in D\\ \beta\in U}}|V_{\alpha\beta}|^2(d_\beta f_\alpha + d_{\overline{\alpha}} f_{\overline{\beta}}) + C^i_{gq}\sum_{\substack{\alpha\in D\\ \beta\in U}}|V_{\alpha\beta}|^2 d_g(f_\alpha + f_{\overline{\beta}}) + C^i_{qg}\sum_{\substack{\alpha\in D\\ \beta\in U}}|V_{\alpha\beta}|^2(d_\beta + d_{\overline{\alpha}})f_g\\
+\;& C^i_{gg}\sum_{\substack{\alpha\in D\\ \beta\in U}}|V_{\alpha\beta}|^2 d_g f_g + C^i_{qq,\mathrm{Fcon},2}\sum_{\substack{\alpha\in D\\ \beta\in U}}|V_{\alpha\beta}|^2\sum_{f\in Q}(d_f f_f + d_{\overline{f}} f_{\overline{f}}) + C^i_{\overline{q}q}\sum_{\substack{\alpha\in D\\ \beta\in U}}|V_{\alpha\beta}|^2(d_{\overline{\beta}} f_\alpha + d_\alpha f_{\overline{\beta}})\\
+\;& C^i_{qq,\mathrm{Fcon},1}\sum_{\substack{\alpha\in D\\ \beta\in U}}|V_{\alpha\beta}|^2(d_\alpha f_\alpha + d_{\overline{\beta}} f_{\overline{\beta}}) + C^i_{\overline{q}q,\mathrm{Fcon}}\sum_{\substack{\alpha\in D\\ \beta\in U}}|V_{\alpha\beta}|^2(d_{\overline{\alpha}} f_\alpha + d_\beta f_{\overline{\beta}})\\
+\;& C^i_{q'q,1}\sum_{f\in Q}\sum_{\substack{\alpha\in D\\ \beta\in U}}|V_{\alpha\beta}|^2(d_f + d_{\overline{f}})(f_\alpha + f_{\overline{\beta}}) + C^i_{q'q,2}\sum_{f\in Q}\sum_{\substack{\alpha\in D\\ \beta\in U}}|V_{\alpha\beta}|^2(d_\beta + d_{\overline{\alpha}})(f_f + f_{\overline{f}})\,.
\end{aligned}
\tag{1.37}
$$

We now move to considering $\mathcal{F}_3^{W^-}$, which reads:

$$
\begin{aligned}
\frac{1}{2}\mathcal{F}_3^{W^-} \;=\;& C_{qq}^3 \sum_{\substack{\alpha\in U\\ \beta\in D}} |V_{\alpha\beta}|^2 (d_\beta f_\alpha - d_{\overline{\alpha}} f_{\overline{\beta}}) + C_{gq}^3 \sum_{\substack{\alpha\in U\\ \beta\in D}} |V_{\alpha\beta}|^2 d_g (f_\alpha - f_{\overline{\beta}}) + C_{qg}^3 \sum_{\substack{\alpha\in U\\ \beta\in D}} |V_{\alpha\beta}|^2 (d_\beta - d_{\overline{\alpha}}) f_g \\
+\;& C_{\overline{q}q}^3 \sum_{\substack{\alpha\in U\\ \beta\in D}} |V_{\alpha\beta}|^2 (d_{\overline{\beta}} f_\alpha - d_{\overline{\alpha}} f_\beta) \\
+\;& C_{qq,\mathrm{Fcon},1}^3 \sum_{\substack{\alpha\in U\\ \beta\in D}} |V_{\alpha\beta}|^2 (d_\alpha f_\alpha - d_{\overline{\beta}} f_{\overline{\beta}}) + C_{\overline{q}q,\mathrm{Fcon}}^3 \sum_{\substack{\alpha\in U\\ \beta\in D}} |V_{\alpha\beta}|^2 (d_{\overline{\alpha}} f_\alpha - d_\beta f_{\overline{\beta}}) \\
+\;& C_{q'q,1}^3 \sum_{f\in Q} \sum_{\substack{\alpha\in U\\ \beta\in D}} |V_{\alpha\beta}|^2 (d_f + d_{\overline{f}})(f_\alpha - f_{\overline{\beta}}) + C_{q'q,2}^3 \sum_{f\in Q} \sum_{\substack{\alpha\in U\\ \beta\in D}} |V_{\alpha\beta}|^2 (d_\beta - d_{\overline{\alpha}})(f_f + f_{\overline{f}})\,.
\end{aligned}
\tag{1.38}
$$

Once again, the exchange of a W^+ is obtained by exchanging $U \leftrightarrow D$:

$$
\begin{aligned}
\frac{1}{2}\mathcal{F}_3^{W^+} \;=\;& C_{qq}^3 \sum_{\substack{\alpha\in D\\ \beta\in U}} |V_{\alpha\beta}|^2 (d_\beta f_\alpha - d_{\overline{\alpha}} f_{\overline{\beta}}) + C_{gq}^3 \sum_{\substack{\alpha\in D\\ \beta\in U}} |V_{\alpha\beta}|^2 d_g (f_\alpha - f_{\overline{\beta}}) + C_{qg}^3 \sum_{\substack{\alpha\in D\\ \beta\in U}} |V_{\alpha\beta}|^2 (d_\beta - d_{\overline{\alpha}}) f_g \\
+\;& C_{\overline{q}q}^3 \sum_{\substack{\alpha\in D\\ \beta\in U}} |V_{\alpha\beta}|^2 (d_{\overline{\beta}} f_\alpha - d_{\overline{\alpha}} f_\beta) \\
+\;& C_{qq,\mathrm{Fcon},1}^3 \sum_{\substack{\alpha\in D\\ \beta\in U}} |V_{\alpha\beta}|^2 (d_\alpha f_\alpha - d_{\overline{\beta}} f_{\overline{\beta}}) + C_{\overline{q}q,\mathrm{Fcon}}^3 \sum_{\substack{\alpha\in D\\ \beta\in U}} |V_{\alpha\beta}|^2 (d_{\overline{\alpha}} f_\alpha - d_\beta f_{\overline{\beta}}) \\
+\;& C_{q'q,1}^3 \sum_{f\in Q} \sum_{\substack{\alpha\in D\\ \beta\in U}} |V_{\alpha\beta}|^2 (d_f + d_{\overline{f}})(f_\alpha - f_{\overline{\beta}}) + C_{q'q,2}^3 \sum_{f\in Q} \sum_{\substack{\alpha\in D\\ \beta\in U}} |V_{\alpha\beta}|^2 (d_\beta - d_{\overline{\alpha}})(f_f + f_{\overline{f}})\,.
\end{aligned}
\tag{1.39}
$$

1.2 Integrating over the phase space

For implementation reasons, we take as a reference for our calculations the cross section differential in x, Q and z integrated over the final-state phase space (bins):

$$
\sigma = \int_{Q_{\min}}^{Q_{\max}} dQ \int_{x_{\min}}^{x_{\max}} dx \int_{z_{\min}}^{z_{\max}} dz \frac{d^3\sigma}{dx dQ dz}\,,
\tag{1.40}
$$

However, sometimes the photon invariant mass Q is replaced by the inelasticity y of the process defined as in terms of the center-of-mass energy $\sqrt{s}$ as:

$$
Q^2 = xys\,.
\tag{1.41}
$$

In this case the integral over the phase space takes the form:

$$
\sigma = \int_{y_{\min}}^{y_{\max}} dy \int_{x_{\min}}^{x_{\max}} dx \int_{z_{\min}}^{z_{\max}} dz \frac{d^3\sigma}{dx dy dz} = \int_{\sqrt{x_{\min} y_{\min} s}}^{\sqrt{x_{\max} y_{\max} s}} dQ \int_{\overline{x}_{\min}}^{\overline{x}_{\max}} dx \int_{z_{\min}}^{z_{\max}} dz \frac{d^3\sigma}{dx dQ dz}\,,
\tag{1.42}
$$

with:

$$
\overline{x}_{\min} = \max\left[x_{\min}, \frac{Q^2}{s y_{\max}}\right] \quad \text{and} \quad \overline{x}_{\max} = \min\left[x_{\max}, \frac{Q^2}{s y_{\min}}\right]\,.
\tag{1.43}
$$

Often, cross sections are measured within a fiducial region defined as:

$$
W = \sqrt{\frac{(1-x)Q^2}{x}} \geq W_{\min}\,, \quad y_{\min} \leq y \leq y_{\max}\,, \quad Q \geq Q_{\min}\,.
\tag{1.44}
$$

These constraints have the effect of reducing the phase space of some bins placed at the edge of the fiducial region. The net effect is that of replacing the x integration bounds in both Eqs. (1.40) and (1.42) with:

$$
x_{\min} \to \overline{x}_{\min} = \max\left[x_{\min}, \frac{Q^2}{s y_{\max}}\right] \quad \text{and} \quad x_{\max} \to \overline{x}_{\max} = \min\left[x_{\max}, \frac{Q^2}{s y_{\min}}, \frac{Q^2}{Q^2 + W_{\min}^2}\right]\,.
\tag{1.45}
$$

Therefore, also the integral in Eq. (1.42) can be recasted in the form of Eq. (1.40). Using Eq. (1.11), we finally have that:

$$\sigma = \sum_j \left[\int_{z_{\min}}^{z_{\max}} dz \left[\int_{Q_{\min}}^{Q_{\max}} dQ \left[\int_{x_{\min}}^{x_{\max}} dx\, \mathbb{K}_i^{\mathrm{SIDIS}}(z|x,Q)\right] \otimes \Gamma_{ij}(z|Q,Q_0)\right] \otimes d_j(z|Q_0)\right] . \tag{1.46}$$

This expression provides the optimal structure for the implementation in APFEL++ in terms of integrations and convolutions. Indeed, upon integration in x, one finds:

$$\int_{x_{\min}}^{x_{\max}} dx\, \mathbb{K}_i^{\mathrm{SIDIS}}(z|x,Q) = c_i(Q; x_{\min}, x_{\max})\mathbb{L}_i(z)\,, \tag{1.47}$$

so that:

$$\begin{aligned}
\int_{Q_{\min}}^{Q_{\max}} dQ \left[\int_{x_{\min}}^{x_{\max}} dx\, \mathbb{K}_i^{\mathrm{SIDIS}}(z|x,Q)\right] \otimes \Gamma_{ij}(z|Q,Q_0) &= \int_{Q_{\min}}^{Q_{\max}} dQ\, c_i(Q; x_{\min}, x_{\max})\mathbb{L}_i(z) \otimes \Gamma_{ij}(z|Q,Q_0)\\
&= \int_{Q_{\min}}^{Q_{\max}} dQ\, \mathbb{M}_j(z|x_{\min}, x_{\max}, Q, Q_0)\\
&= \mathbb{N}_j(z|x_{\min}, x_{\max}, Q_{\min}, Q_{\max}, Q_0)
\end{aligned} \tag{1.48}$$

and finally:

$$\sigma = \sum_j \int_{z_{\min}}^{z_{\max}} dz\, \mathbb{N}_j(z|x_{\min}, x_{\max}, Q_{\min}, Q_{\max}, Q_0) \otimes d_j(z|Q_0) = \sum_j \mathbb{O}_j(z_{\min}, z_{\max}, x_{\min}, x_{\max}, Q_{\min}, Q_{\max}, Q_0)\,. \tag{1.49}$$

2 Computation of the single-inclusive e^+e^- annihilation cross section

From the point of view of the numerical computation, the single-inclusive e^+e^- annihilation (SIA) cross section can be regarded as a simplified version of the SIDIS one. Specifically, for SIA the variable x is just absent and there is no integration over the scale Q because $Q = \sqrt{s}$ with $\sqrt{s}$ fixed and determined by the collision energy of the lepton pair. In addition, in all cases considered in our analysis, data are delivered for specific values of the variable z such that no integration over z is required. Finally, the observable is computed as:

$$\frac{1}{\sigma}\frac{d\sigma}{dz} = \sum_j \mathbb{K}_i^{\mathrm{SIA}}(z|Q) \otimes \Gamma_{ij}(z|Q,Q_0) \otimes d_j(z|Q_0)\,. \tag{2.1}$$

This equation closely resembles Eq. (1.11) but the $\mathbb{K}_i^{\mathrm{SIA}}$ functions have a simpler structure and, up to a factor σ_0/σ, where σ_0 is the leading-order total cross section, are already computed by APFEL up to $\mathcal{O}(\alpha_s^2)$.

References

[1] L. Bonino, T. Gehrmann, and G. Stagnitto, “Semi-inclusive deep-inelastic scattering at NNLO in QCD,” 1 2024.

[2] L. Bonino, T. Gehrmann, M. Löchner, K. Schönwald, and G. Stagnitto, “Polarized semi-inclusive deep-inelastic scattering at NNLO in QCD,” 4 2024.

[3] L. Bonino, T. Gehrmann, M. Löchner, K. Schönwald, and G. Stagnitto, “Neutral and Charged Current Semi-Inclusive Deep-Inelastic Scattering at NNLO QCD,” 6 2025.